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The Next Generation Transit Survey: high-precision targeted-mode operations

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ABSTRACT

The Next Generation Transit Survey (NGTS) operates an array of twelve fully robotic, independent 20-cm telescopes equipped with red-sensitive cameras at Cerro Paranal, Chile. Alongside its exoplanet survey, NGTS provides high-precision photometry for targeted observations of transiting planets, stellar activity, white dwarfs, and young stellar associations. Follow-up of candidates from the Transiting Exoplanet Survey Satellite (*TESS*), together with coordinated observations with other facilities, requires rapid data reduction. We describe the high-precision targeted observing mode of NGTS and the quick-look photometric pipeline developed to process these observations using the infrastructure available at Paranal. The pipeline identifies contamination from laser guide stars and satellite trails, estimates spatially varying sky backgrounds, refines astrometric solutions, extracts aperture photometry, and removes common systematic trends. It produces publication-quality light curves for individual targets, together with diagnostic information for assessing photometric quality and possible transit signals. We detail the observing strategy, reduction procedures, and data products, and assess the photometric and tracking performance. The resulting light curves achieve precision comparable to the NGTS noise model, supporting timely follow-up and coordinated observing campaigns.

Key words: methods: data analysis – techniques: image processing – techniques: photometric

1 INTRODUCTION

The Next Generation Transit Survey (NGTS; Wheatley et al. 2018) operates an array of twelve telescopes at the European Southern Observatory (ESO) Paranal Observatory. Each 20-cm telescope is designed for high-precision photometry, achieving approximately 400 ppm (0.04 per cent) in 30 min for bright stars. Combining observations from multiple telescopes can improve this precision to approximately 100 ppm (0.01 per cent) in 30 min (Bryant et al. 2020a). Each telescope has a field of view of 8 $\deg^2$, providing a large pool of potential reference stars for differential photometry. A custom filter spanning 520–890 nm is used for observations of exoplanet transits. Images are acquired at a cadence of approximately 13 s using modified Andor Technology Ltd iKon-L cameras, built around red-sensitive, deep-depletion e2v CCDs with an illuminated area of 2048×2048 pixels.

NGTS achieved first light in 2015 and began full survey operations in 2016 (Wheatley et al. 2018). The original survey strategy assigned each telescope a separate field to monitor over successive nights while it remained observable. The stability of the telescopes, together with guiding to approximately 0.1 pixels relative to a reference image, provides stable time-series photometry for identifying exoplanet candidates. One of the survey's aims was to expand the population of Neptune-sized and smaller planets discovered around Sun-like stars. Its discoveries include hot Jupiters and planets in the Neptune desert.[1]

The data-handling strategy was designed around the requirements of the survey. Each telescope operates independently, although several telescopes can observe the same field of approximately $2.8° \times 2.8°$. For a field not previously observed with a particular camera, the first image becomes the autoguider reference image. Subsequent images of that field with the same camera are guided relative to this reference using the DONUTS algorithm (McCormac et al. 2013). Images are collected in observing blocks called 'actions', typically comprising observations of one field with one camera over a night, and each action is assigned a unique identification number. Images are stored on each telescope's data-acquisition server before being compressed and transferred to central storage at Paranal. They are subsequently shipped on hard disks to the University of Warwick for permanent storage. In the original survey workflow, data accumulated over one or two years before being processed in bulk, as described by Wheatley et al. (2018).

The observing strategy has evolved to support follow-up of exoplanet candidates from the Transiting Exoplanet Survey Satellite (*TESS*; Ricker et al. 2015). Survey discoveries continue alongside targeted observations of promising candidates. In this targeted observing mode, NGTS has obtained precise multicamera transit photometry of *TESS* Objects of Interest (TOIs; e.g. Bryant et al. 2020a; Armstrong et al. 2020; Barragán et al. 2022) and recovered the orbital periods of warm Jupiters identified as *TESS* single-transit candidates (Gill et al. 2020b). Its auxiliary science programme also includes observations complementary to the James Webb Space Telescope (*JWST*; e.g. WASP-39; Ahrer et al. 2023), young stellar associations (Smith et al. 2023), white dwarf pulsators (Chote et al. 2021), and eclipsing binaries (Gill et al. 2020a).

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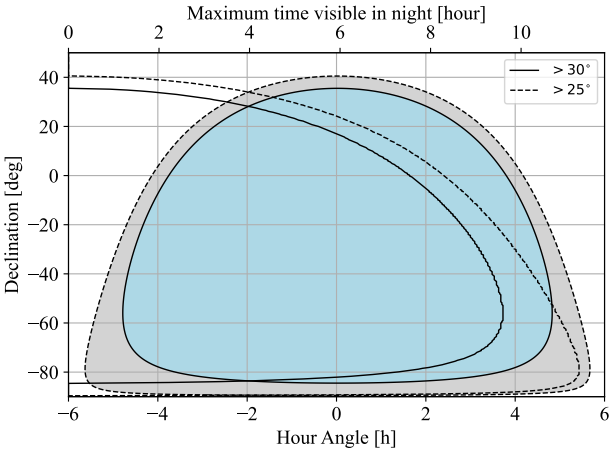


Figure 1. Target visibility from NGTS. Grey contours bounded by dashed black lines show the region in which a target can be observed at elevations above 25°. Blue contours bounded by solid black lines show the corresponding region for elevations above 30°. The top axis shows the maximum time a target is visible above 25° (dashed black line) and 30° (solid black line).

Targeted observing requires data to be processed and analysed within a few days, motivating the development of the NGTS quick-look pipeline. This pipeline produces publication-quality light curves and has also informed improvements to the main survey pipeline described by Wheatley et al. (2018). Here we provide a high-level description of the observing strategy and reduction workflow. Section 2 describes targeted operations and complementary facilities. Section 3 covers image processing, including the refinement of the world coordinate system (WCS) in Section 3.5. Sections 4 and 5 describe photometric extraction and data products, respectively. Section 6 assesses performance, and Section 7 presents our conclusions.

2 TARGETED OBSERVING MODE

NGTS uses two principal strategies in *targeted observing mode*. The first is to observe known or suspected transiting exoplanets with one or more cameras. These observations include TOI follow-up (e.g. Bryant et al. 2020a; Armstrong et al. 2020; Barragán et al. 2022), monitoring transit timing variations (e.g. Kaye et al. 2022; Leleu et al. 2024), supporting radial-velocity campaigns, including simultaneous Rossiter–McLaughlin observations with ESPRESSO (Doyle et al. 2022, 2023), and determining the orbital periods of cool giant-planet candidates with multiple period aliases in *TESS* data (e.g. Gill et al. 2020a,b). The second strategy is to monitor a star for a short interval each night (typically 20–30 min) at optimal airmass, for a campaign lasting from days to months. These observations can constrain stellar activity or complement spectroscopic monitoring to help distinguish activity-induced signals from planetary radial-velocity signals (Ahrer et al. 2023).

The following sections describe the capabilities, scheduling, and scientific applications of targeted observing with NGTS.

2.1 Observational capabilities

NGTS is located alongside the Very Large Telescope (VLT) at ESO's Paranal Observatory in Chile, enabling simultaneous photometry to support spectroscopy with ESPRESSO (Pepe et al. 2010). Paranal offers a high fraction of photometric nights and dry conditions that reduce variability in telluric water absorption at red wavelengths. NGTS has good access to the southern sky (Fig. 1) and can monitor targets with declinations as far north as +20°. It has also supported coordinated campaigns at multiple longitudes to observe long-duration transits of long-period planets. NGTS routinely observes TOIs, community TOIs, and candidates identified by dedicated searches for long-period planets. Cerro Paranal is well placed for observations of the southern ecliptic hemisphere and provides more limited access to the northern ecliptic hemisphere. NGTS can also support candidate validation, simultaneous observations, and preparatory surveys in the southern long-duration observation field of *PLATO* (LOPS2). These complementary programmes are discussed in Section 2.4.

2.2 Precision

Fig. 2 compares noise models for NGTS and *TESS*. A single NGTS telescope is expected to provide photometric precision comparable to *TESS* for stars fainter than $G = 12$. Blending and scattered light, which are not included in the *TESS* noise model, can reduce the precision of *TESS* observations relative to this prediction. For stars brighter than $G = 12$, a single NGTS telescope provides a precision of approximately 400 ppm (0.04%) in 30 min, limited primarily by atmospheric scintillation. When correlated noise is negligible, averaging exposures further improves the precision.

As an example, the right-hand panel of Fig. 2 shows the discovery light curves of NGTS-12 b (Bryant et al. 2020b), a low-density planet with an orbital period of 7.5 d around a $G = 12.1$ host star. The NGTS and *TESS* observing times are comparable (660 and 510 h, respectively), and the single-telescope NGTS light curve has a lower rms scatter in this example. The *TESS* transit is slightly shallower because of blending with neighbouring stars, which also contributes to the photometric noise.

For stars brighter than $G = 12$, scintillation noise can be reduced by combining simultaneous observations from multiple NGTS telescopes. The noise model in Fig. 2 predicts a precision of approximately 100 ppm in 30 min when all twelve telescopes are combined. This approach is supported by observations of WASP-166 b obtained simultaneously with *TESS* and nine NGTS telescopes (Bryant et al. 2020a), shown in Fig. 3. For this $G = 9.3$ star, the combined NGTS light curve has approximately the same precision as *TESS*.

A larger-aperture telescope can also reduce scintillation noise. However, for bright targets, the wide field of view of NGTS helps provide sufficient reference stars for precise differential photometry. This can be more difficult with the narrower fields of view of 1-m-class telescopes. Each NGTS telescope has a field of view of 8 deg^2, enabling sub-millimagnitude relative photometry for bright *TESS* candidates down to $G \sim 6$. Combining all 12 telescopes on a single target provides the collecting area of a 0.7-m telescope and the scintillation noise of a 1.3-m telescope (O'Brien et al. 2022). This configuration can achieve approximately 100 ppm in 30 min for bright stars (see Figs 2 and 3, and Bryant et al. 2020a).

The optical and mechanical design of NGTS, together with precise polar alignment, limits image motion caused by field rotation to less than one pixel near the edges of the CCD over a night. Baffles reduce contamination from scattered moonlight, although the lunar cycle remains prominent in the sky background (Fig. 4). This is particularly relevant to long-term stellar-activity monitoring. Sky background also increases with airmass: an observation at airmass 2 can have almost twice the sky background of one at airmass 1, before

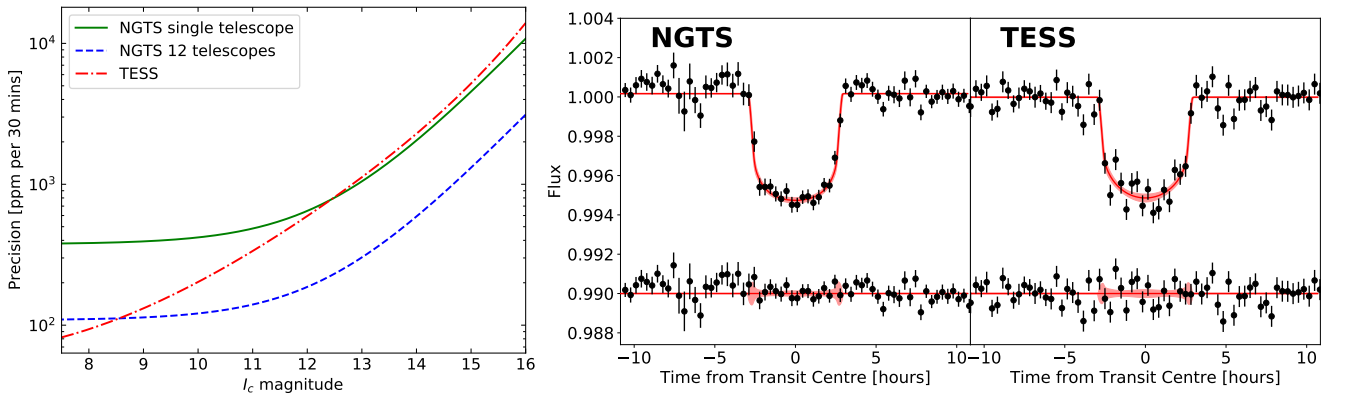


Figure 2. Left: comparison of photometric noise models for a single NGTS telescope (solid green line), 12 NGTS telescopes (dashed blue line), and *TESS* (dot–dashed red line). A single NGTS telescope and *TESS* have similar modelled photometric precision for stars of 12th magnitude and fainter. For brighter stars, a single NGTS telescope is limited by atmospheric scintillation to approximately 400 ppm in 30 min. Combining 12 telescopes reduces this to approximately 100 ppm, comparable to *TESS* for stars as bright as 9th magnitude (see Fig. 3). **Right:** phase-folded NGTS and *TESS* discovery light curves for NGTS-12 (G = 12.1), with the best-fitting models shown as solid red lines (Bryant et al. 2020b). The NGTS light curve comprises 660 h of observations from a single camera (rms = 400 ppm). The *TESS* light curve comprises 510 h of observations from Sector 10 (rms = 450 ppm). Blending in the larger *TESS* pixels contributes to the lower photometric precision and slightly shallower transit depth.

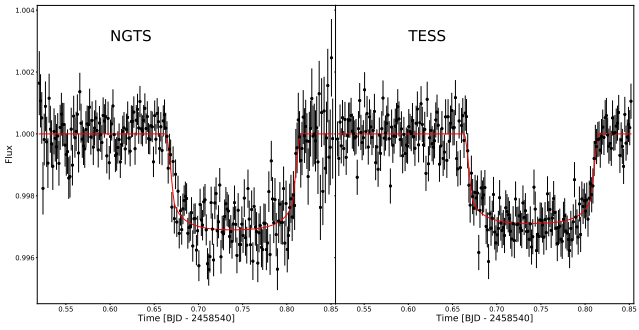


Figure 3. A single transit of WASP-166 b across its bright host star (G = 9.3) on 2019 February 25, observed with NGTS (left) and *TESS* (right). The NGTS light curve combines simultaneous observations from nine telescopes. The data are binned to 2 min, and the solid red line shows the best-fitting transit model. Figure from Bryant et al. (2020a).

additional contributions from scattered moonlight and high-altitude clouds.

2.3 Scheduling observations

Each of the 12 telescopes is fully robotic and can observe at hour angles within ±5.3 h and declinations between −90° and +20°. The telescopes usually operate independently, allowing up to 12 separate observing sequences. When greater precision is required, multiple telescopes can observe the same field and their photometry can be combined. Two features of NGTS support this approach. First, each telescope uses a long-term reference image (typically the first image of a field) and the DONUTS algorithm (McCormac et al. 2013) to maintain subpixel guiding stability over many nights, reducing correlated noise associated with image motion. Secondly, the separation between telescopes makes their atmospheric scintillation effectively uncorrelated. These properties allow averaging both within and across telescopes, reaching approximately 100 ppm in 30 min when multiple telescopes are combined (Bryant et al. 2020a).

A typical night for each telescope comprises evening flats, a focus sweep, science observing blocks, morning flats, and bias and dark exposures after the roof has closed. Each phase is defined by an NGTS action, with parameters specifying its type, start and end times, and any action-specific settings, such as the target count level for flats or the range of a focus sweep. The NGTS scheduler enters these parameters into a MySQL database table. Each telescope's control system (PALADIN) retrieves actions in real time to execute the nightly sequence. Actions are scheduled according to solar elevation: for ex-

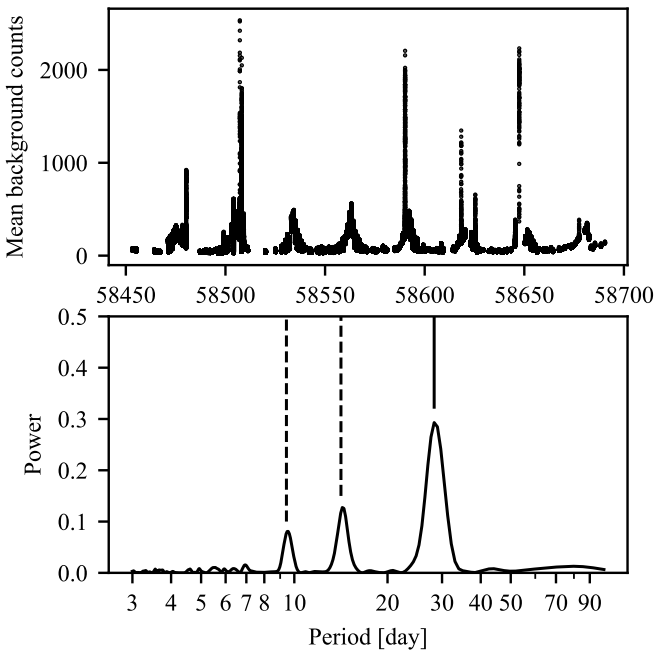


Figure 4. Upper panel: mean background counts over a season of observations of a field centred at RA = 10^h19^m, Dec = $-33°35'$. **Lower panel:** Lomb–Scargle power spectrum of the background time series. The solid black line marks the lunar-cycle period, and the dashed black lines mark its first two aliases.

ample, the roof opens at +1.0°, dusk flats begin at −4.5°, and science observations begin at −15°.

The scheduler distinguishes between time-critical and survey observations. Time-critical observations have user-specified start and end times. Survey targets are eligible whenever they satisfy the telescope's observability constraints, such as minimum elevation and angular separation from the Moon. Their observing windows are then determined by the visibility and priorities of competing targets. The working groups allocate the available observing time among their projects and agree on target priorities; the scheduler applies these priorities without managing the scientific allocation itself.

The scheduler aims to observe the highest-priority eligible target on each telescope at any given time. Working groups assign integer priorities from 0 (highest) to 10 (lowest). Routine priorities use odd integers, while priority 0 is reserved for exceptional targets. Even integers allow new requests to be inserted between existing priority levels at short notice. Time-critical observations are assigned higher priorities than survey observations.

The observing window, defined by solar elevations below −15°, is divided into 1-min steps. In an initial pass, the scheduler combines each target's priority and airmass to evaluate its *importance* at every step. A second pass identifies continuous blocks during which a target has the highest importance and sets their start and end times. Including airmass allows survey targets of equal priority to hand over at the optimal airmass, reducing the time spent observing at high airmass.

Each telescope's schedule is prepared in advance and comprises time-critical observations and lower-priority survey fields. Where possible, the same cameras are used for a target on successive nights to avoid systematic offsets between camera photometric zero-points.

Default limits on elevation, lunar separation, and twilight observing are preconfigured. Users can adjust these within predefined safety limits when required.

At the end of the night, images are compressed using BZIP2 (Seward 1998) and moved to centralised storage at Cerro Paranal. Transfer to permanent storage at the University of Warwick can take several months because the data are transported in bulk on removable 6-TB hard disks.

2.4 Complementary projects

The capabilities of NGTS complement those of other observatories. The following examples illustrate its role in exoplanet discovery, characterisation, and monitoring.

2.4.1 *TESS*

The Transiting Exoplanet Survey Satellite (*TESS*; Ricker et al. 2015) is a NASA Explorer-class spacecraft conducting a wide-field search for transiting exoplanets around bright stars. *TESS* began science operations in July 2018 and surveyed approximately 80% of the sky during its two-year primary mission. Its extended missions revisit large areas of the sky. A primary science goal is to enable mass measurements for 50 transiting exoplanets with radii $R_{\rm p} < 4R_{\oplus}$, where $R_{\oplus}$ is the Earth's radius.

The stability of space-based observations and the absence of atmospheric scintillation and a diurnal cycle make *TESS* well suited to detecting transiting exoplanet candidates. However, a single sector covers approximately 27 d, limiting the number of observed transits for longer-period planets. Candidates that transit only once or twice may therefore require extensive follow-up to determine their orbital periods. Resolving these period ambiguities is a major science objective of NGTS (Gill et al. 2020a,b, 2022, 2024). In addition, the large *TESS* pixels (21 arcsec) can lead to substantial blending (Sullivan et al. 2015).

The finer NGTS pixel scale (5 arcsec pixel^{-1}) resolves many sources blended in *TESS* photometry. Its 13-s cadence also supports detailed studies of stellar variability, including variability in white dwarfs (e.g. Chote et al. 2021). For comparison, *TESS* full-frame images have a cadence of 200 s from Cycle 5 onwards, with 20-s and 120-s products available for selected targets.[2] As shown in Fig. 2, a single NGTS telescope and *TESS* have comparable modelled precision for stars fainter than $G = 12$. NGTS members participate in the *TESS* Follow-up Observing Program (TFOP), particularly Sub-Group 1 (photometric follow-up) and Sub-Group 4 (radial-velocity follow-up).

High-precision ground-based differential photometry requires suitable reference stars within the instrument's field of view. For bright targets, finding reference stars of comparable brightness can be difficult in a small field. The wide field of NGTS helps provide a suitable reference-star ensemble.

NGTS has used its multi-telescope mode to provide sub-millimagnitude photometry for bright TOIs (Armstrong et al. 2020; Jenkins 2019; Nielsen et al. 2020; Osborn et al. 2021; Kaye et al. 2022; Hawthorn et al. 2023). It also conducts multi-year monitoring of bright transiting systems to measure transit timing variations (TTVs) associated with planet–planet interactions or orbital evolution in short-period systems. Examples include a large TTV of the low-density, 542-d-period planet HIP 41378 f (Bryant et al. 2021) and TTVs in the TOI-270 system (Kaye et al. 2022).

2.4.2 *CHEOPS*

CHEOPS (CHaracterising ExOPlanet Satellite; Benz et al. 2021) is an ESA S-class mission launched in December 2019. Its 0.3-m telescope performs high-precision photometry to measure planetary radii. *CHEOPS* is a dedicated follow-up mission that observes a single star at a time using a heavily defocused image on a frame-transfer CCD.

NGTS and *CHEOPS* offer complementary capabilities. NGTS supports long-term monitoring, for example to eliminate period aliases of *TESS* single-transit candidates, while the greater photometric precision of *CHEOPS* enables shallower transits and more precise radii to be measured. The twelve NGTS telescopes can be divided among several targets, while *CHEOPS* observes targets individually. NGTS also provides access to southern targets outside the observing constraints of *CHEOPS*.

The six-planet TOI-178 system illustrates this complementarity (Leleu et al. 2021). *TESS* initially identified three candidates with periods of 6.55, 10.35, and 9.96 d (Leleu et al. 2019). Targeted NGTS observations detected only one of two predicted transits of the 10.35-d candidate. Together with *CHEOPS* observations, these data established a period of 20.7 d, with one of the apparent 10.35-d transits in the *TESS* data attributed to an additional planet with a period of 15.2 d. *CHEOPS* also confirmed two inner planets with periods of 1.9 and 3.2 d. The outer five planets form a 2:4:6:9:12 chain of Laplace resonances.

[2] https://heasarc.gsfc.nasa.gov/docs/tess/data-products.html

2.4.3 ESPRESSO

ESPRESSO (Echelle Spectrograph for Rocky Exoplanets and Stable Spectroscopic Observations; Pepe et al. 2010) is an ultra-stable spectrograph fed by the VLT at Paranal. It was designed for radial-velocity precision of $10\,\mathrm{cm\,s^{-1}}$ to enable the detection of Earth-like planets. It can receive light from a single VLT Unit Telescope or from all four simultaneously, providing access to fainter targets than instruments on smaller telescopes, such as HARPS (Pepe et al. 2002). This supports radial-velocity monitoring of optically faint targets, including M dwarfs.

NGTS observations support ESPRESSO programmes by identifying false positives among *TESS* candidates, including blended eclipsing binaries, and by providing simultaneous high-precision photometry to constrain stellar activity. During transits, such observations can identify starspot crossings and help assess the effects of stellar heterogeneity on Rossiter–McLaughlin measurements and transmission spectroscopy. The location of both instruments at Paranal facilitates simultaneous observations, as demonstrated by Doyle et al. (2022, 2023).

2.4.4 PLATO

PLATO is an ESA M-class mission designed to characterise stars using asteroseismology and detect transiting exoplanets as small as Earth around bright Sun-like stars, including planets in the habitable zone (Rauer et al. 2014).

PLATO combines observations from multiple small telescopes with overlapping fields of view, an approach also used by NGTS to improve photometric precision. Potential contributions from NGTS include preparatory surveys of *PLATO* fields, period confirmation and TTV monitoring after *PLATO* moves to another field, and observations of eclipsing white dwarfs, long-period variable stars, and benchmark eclipsing binaries.

3 IMAGE PROCESSING

The deep-depletion CCDs on NGTS have an illuminated area of 2048×2048 pixels. Unilluminated prescan and overscan regions (the first and last 20 columns, respectively) increase the raw image size to 2048×2088 pixels. These regions allow us to measure frame-to-frame variations in the bias level. We subtract the median overscan count on a row-by-row basis and crop the image to remove the prescan and overscan regions. The resulting images are used as our reduced science images. Unlike the main NGTS pipeline (Wheatley et al. 2018), we do not apply flat-field, dark-current or further bias corrections. We find that these additional corrections do not improve the photometric precision and are therefore unnecessary for producing publication-quality light curves.

3.1 Laser detection

The European Southern Observatory Very Large Telescope (VLT) comprises four 8-m telescopes adjacent to NGTS at Cerro Paranal and hosts the 4 Laser Guide Star Facility (Calia et al. 2014). This system projects four laser beams into the atmosphere, exciting sodium atoms to create artificial guide stars for adaptive-optics correction of atmospheric turbulence. Occasionally, these lasers traverse the field of view of an NGTS telescope. Similar laser systems planned for the other three 8-m telescopes and the European Southern Observatory Extremely Large Telescope (ELT; Thatte et al. 2016) could increase the frequency of these events.

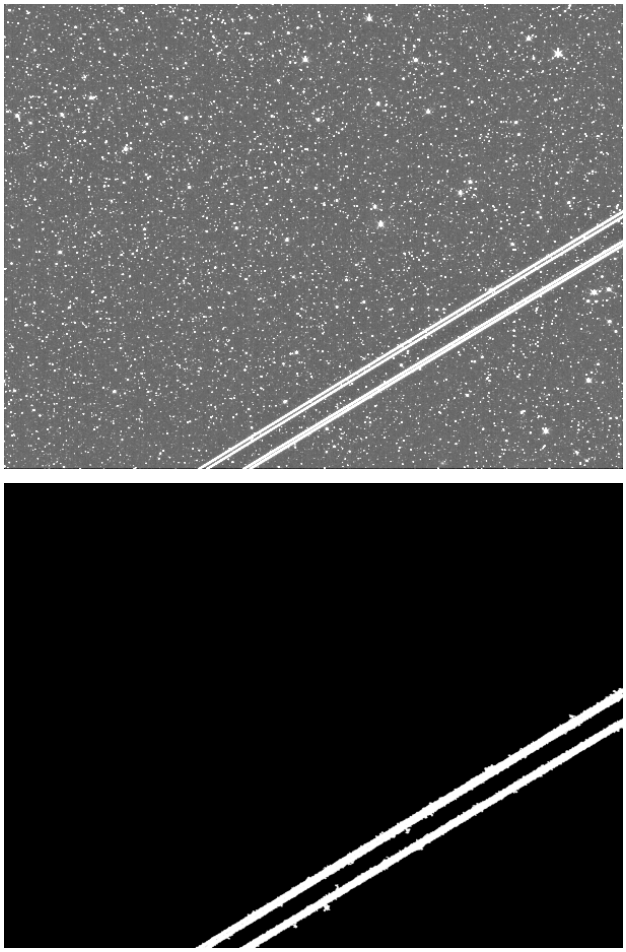


Figure 5. Upper panel: raw image of a field centred at RA = $14^{\mathrm{h}}\,16^{\mathrm{m}}$, Dec = $-30^{\circ}\,56'$, with the VLT laser visible in the bottom-right corner. Lower panel: binary mask identifying pixels affected by the laser.

We identify images containing laser trails and flag the affected pixels. We first flag all pixels with counts exceeding 1000 ADU, then recursively flag neighbouring pixels whose counts exceed the image median by more than 1σ, where σ measures the pixel-value scatter. Adjacent flagged pixels are grouped, and each group is checked to determine whether it touches two edges of the frame. If so, we attribute the group to a laser trail and flag all sources within 50 pixels of the grouped pixels. This algorithm, illustrated in Fig. 5, distinguishes laser artefacts from bright stellar sources. Identifying these artefacts is important for autoguiding with the DONUTS algorithm and for flagging affected photometric measurements.

3.2 Satellites

Satellites can appear as point-like sources or extended streaks, depending on their apparent motion, shape, size and orientation relative to the Sun. We do not attempt to identify slow-moving, point-like satellites because distinguishing them from stars in individual images is difficult; tracking them across successive frames is beyond the scope of this pipeline. For satellites producing streaks, we use the

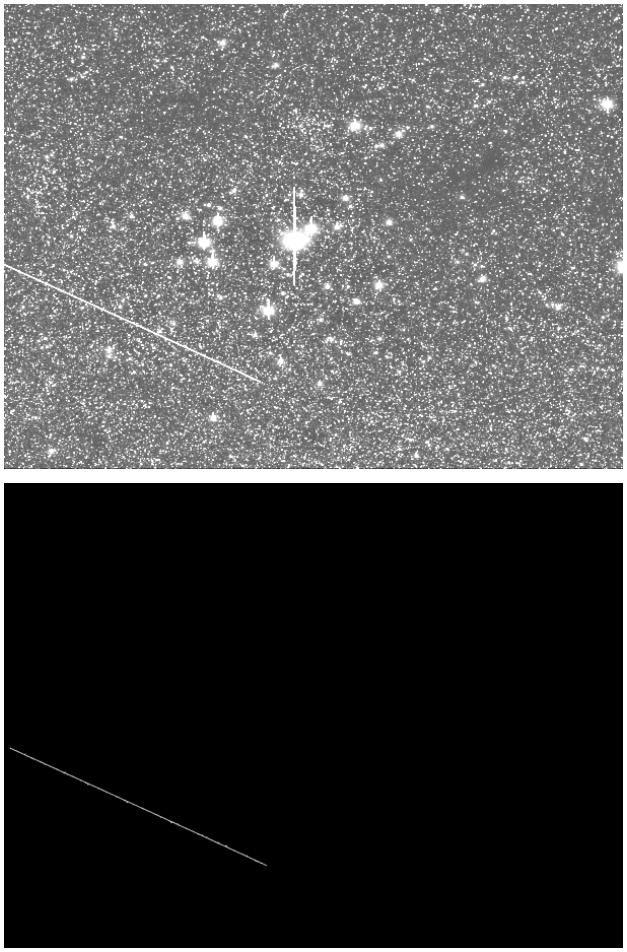


Figure 6. Upper panel: raw image of a field centred at RA = $10^h\,42^m$, Dec = $-64°\,25'$ (IC 2602), with a prominent satellite streak. Lower panel: binary mask produced by ASTRIDE, identifying pixels affected by the satellite.

ASTRIDE package (Kim 2016), which identifies streaks through contour tracing. All pixels within 50 pixels of a detected satellite trail are flagged; an example is shown in Fig. 6. The algorithm can produce false detections, particularly around bright sources, so this flag is used with caution.

3.3 Estimating the sky background

Like many ground-based observations, NGTS images have a spatially varying sky background. Its structure reflects instrumental effects, such as vignetting and dust in the optical path, together with diffuse sky emission and scattered light. Variations in the background produce spatial variations in the photometric uncertainty. We therefore require spatially resolved background maps to measure stellar fluxes and their uncertainties.

Iterative sigma clipping identifies pixels more than 7σ above the median, which may be affected by cosmic rays, satellites or the VLT adaptive-optics lasers (Section 3.1). We then generate a spatially varying background map using SEP (Bertin & Arnouts 1996; Barbary 2016), with background boxes of 128×128 pixels and a filter spanning 3×3 background boxes. Subtracting this map yields the background-subtracted science image used for photometry (Fig. 7).

3.4 Source detection

Accurate stellar positions are needed to mask sources during background estimation and to determine the astrometric solution. We use the SEP function `extract` (Barbary 2016), which is based on SExtractor (Bertin & Arnouts 1996). Appropriate filtering kernels are needed to detect sources in defocused images, where the point-spread function (PSF) changes from an approximately Gaussian profile with a full width at half-maximum (FWHM) of ~ 2 pixels to a broad annulus. The defocused PSF is much broader than the atmospheric seeing disc at Cerro Paranal and is therefore relatively insensitive to seeing variations. We imaged a sparse field at defocus offsets ranging from 0 mm (in focus) to 0.6 mm (maximum defocus) and measured the mean PSF of suitable stars (left column of Fig. 8). Below 0.2 mm, we fitted Gaussian profiles at defocus offsets of 0 and 0.1 mm to construct the filtering kernels (middle column). At offsets of 0.2 mm and above, we fitted annular profiles, determining the inner radius and the FWHM of the ring. Kernels for intermediate defocus values are interpolated from these kernels, allowing stellar positions to be measured from filtered images (right column).

This process yields source positions, fluxes and elliptical PSF shape parameters for each NGTS image. These measurements are used to construct a binary mask of pixels containing stellar flux for background estimation (Section 3.3) and to refine the world coordinate system (WCS; Section 3.5).

3.5 Determining the world coordinate system

We derive an initial WCS solution for each image using astrometry.net (Lang et al. 2010). We use a third-degree Simple Imaging Polynomial (SIP) model to describe distortion across the image (Shupe et al. 2005) and allow astrometry.net to refit the polynomial coefficients. Cross-matching sources with the *TESS* Input Catalog (TIC), however, reveals residual distortions near the edges of the field, with positional offsets reaching several tens of pixels (middle panel of Fig. 9).

We refine the SIP coefficients using a catalogue of stars from TIC v8.0 (Stassun et al. 2019) that lie within the frame, have valid *Gaia* DR2 identifiers and satisfy $10 < T_{\rm mag} < 15$, where $T_{\rm mag}$ is the *TESS* magnitude. Pixel positions of detected sources (Section 3.4) are converted to right ascension and declination using the initial WCS and cross-matched with the catalogue. The SIP coefficients are then refitted to account for the residual distortion. Measured source brightnesses are used to reject potential mismatches with inconsistent fluxes. We repeat this process until the change in the median improvement is less than 0.01 pixels in both dimensions, or until 20 iterations have been completed; most frames require 3–7 iterations. The right-hand panel of Fig. 9 shows the resulting residuals, and Fig. 10 compares the positional offsets before and after refinement. With astrometry.net alone, most stars have offsets of 1–10 pixels, compared with a typical in-focus PSF width of ~ 3 pixels. After refinement, most positions agree to within ~ 0.1 pixels.

4 PHOTOMETRY

4.1 Target photometry catalogue

Each autoguider reference image has an associated photometry catalogue containing TIC v8.0 sources that fall on the detector. We extract photometry for all sources with $T_{\rm mag} < 16$. This limit may be relaxed when observing a fainter target and selecting its comparison

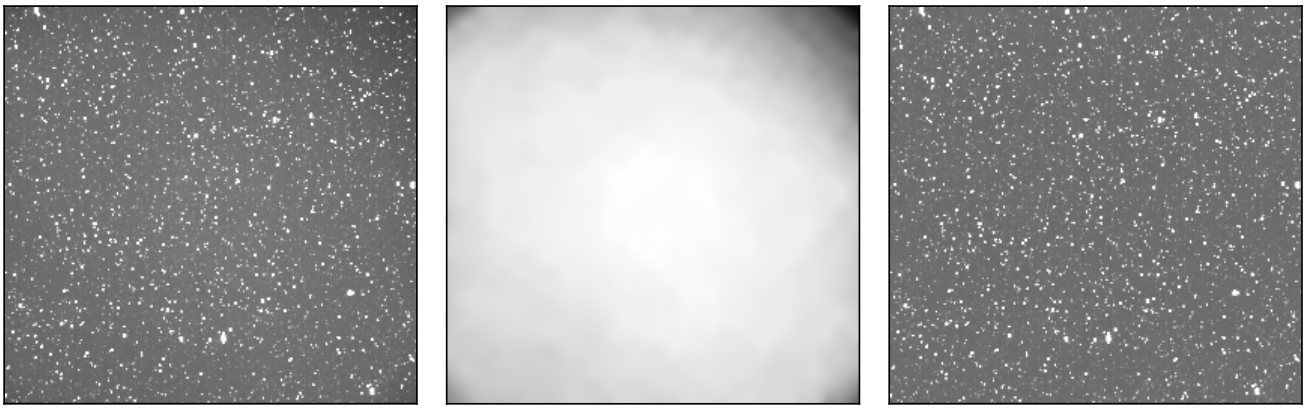


Figure 7. Left: raw image of a field centred at RA = $21^{h}\,41^{m}$, Dec = $-36^{\circ}\,33'$. Centre: sky background estimated using SEP. Right: background-subtracted image used for aperture photometry.

Table 1. Aperture radii used for photometry at different defocus offsets, D.

Image defocus, D (mm)	Aperture radius (pixels)
$0 \leq D < 0.1$	3.0
$0.1 \leq D < 0.2$	3.5
$0.2 \leq D < 0.3$	4
$0.3 \leq D < 0.4$	4.5
$0.4 \leq D < 0.5$	5
$0.5 \leq D < 0.6$	6
$D > 0.6$	7

stars. Sources brighter than $T_{\mathrm{mag}} = 9$ are retained for focused observations, although they are typically saturated.

4.2 Aperture photometry of each frame

We perform aperture photometry on the background-subtracted frame using the SEP routine `sum_circle`. We extract photometry for every star in the target photometry catalogue (Section 4.1), using an aperture radius appropriate to the image defocus (Table 1). These radii were determined empirically from images taken at different defocus offsets (Section 3.4 and Fig. 8) and enlarged by 0.5 pixels to accommodate seeing-induced PSF broadening. Fainter stars can benefit from a 2-pixel radius aperture when the photometry is background-limited. We nevertheless adopt a minimum radius of 3 pixels for stability and consistency with the pipeline presented by Wheatley et al. (2018).

Apertures are centred using each frame's improved WCS solution (Section 3.5). The exact overlap between each aperture and the pixels is used to determine the flux and its uncertainty, with predetermined gain values for each camera. We also report the maximum value among all pixels intersecting each aperture to help identify saturation or contamination by nearby bright stars.

4.3 Time conversions

We use the ASTROPY Python package (Astropy Collaboration et al. 2013, 2018, 2022) for time conversions. For each image, we obtain the modified Julian date (MJD) at mid-exposure and calculate the barycentric and heliocentric light-travel-time corrections for each object in the target photometry catalogue. We report heliocentric Julian dates on the Coordinated Universal Time scale ($\mathrm{HJD_{UTC}}$) and barycentric Julian dates on the Barycentric Dynamical Time scale ($\mathrm{BJD_{TDB}}$).

4.4 Measuring the point-spread function

Temporal variations in the PSF can introduce systematic features in aperture photometry, including spurious transit-like signals. We characterise the PSF using three methods. First, we use the source-detection measurements from SEP to estimate the FWHM of each detected source. For each frame, we report the median FWHM across all detected sources and the median for sources within the central 1024×1024 pixels. Secondly, we partition the frame into a 3×3 grid. Sources within each cell are stacked and resampled to construct a supersampled PSF, from which we estimate the width, length and orientation of a representative ellipse. Finally, we measure the FWHM from a rolling stack of the last 10 frames, using SEXTRACTOR (Bertin & Arnouts 1996), and report the mean FWHM for each frame. These measurements support quality assessment and detrending of the observations.

4.5 Centroiding

Accurate stellar positions on the CCD help identify systematic errors arising from non-uniform pixel response. Despite the precise subpixel guiding of NGTS, occasional guiding errors can introduce systematic photometric trends. We provide positional information for each target to help identify and correct these effects. We report positions predicted by the improved WCS solution (Section 3.5) and positions measured using the SEP function `winpos`, which iteratively recentres a window around each star. The standard window size is 9 pixels, with adjustments for defocused observations.

5 DATA PRODUCTS

5.1 Low-level products

Low-level data products are archived as compressed (BZIP2) FITS files in one directory per action, with one file for each data product, totalling 12 files per directory. The first file contains the barycentric

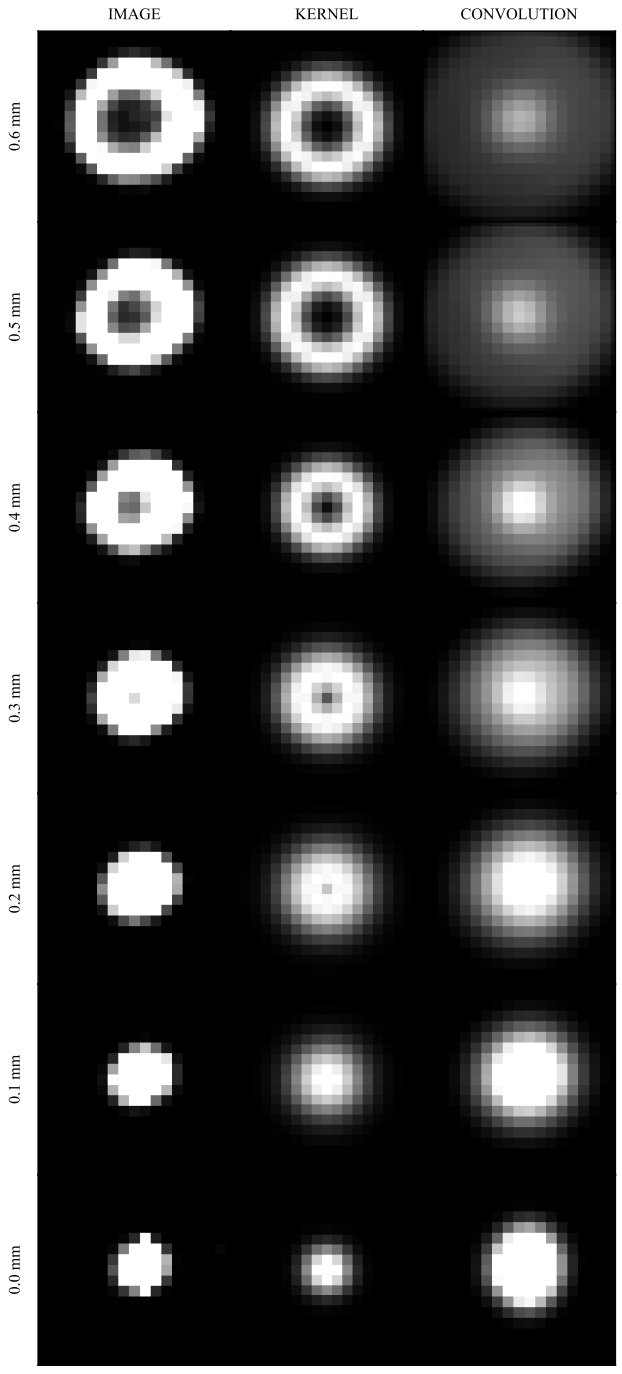


Figure 8. Point-spread functions of sources in NGTS images at different defocus offsets (in mm; left column), the corresponding kernels used for source detection (middle column), and the convolution of each point-spread function with its kernel (right column).

mid-exposure time for each source in each frame. These times are stored in a two-dimensional FITS array of 32-bit integers, as the number of seconds elapsed since 2014-01-01T00:00:00. Three files store the flux in the optimal aperture, its formal uncertainty, and the sky background flux for each source. Each is stored in a two-dimensional FITS array of 32-bit floating-point numbers. The WCS aperture positions (Section 3.5) and centroid positions (Section 4.5) are stored separately for each image dimension, in two-dimensional FITS arrays of 16-bit unsigned integers. An observation summary file contains a FITS table in extension 1, providing auxiliary information such as airmass, humidity, and the angular separation and brightness of the Moon.

5.2 Detrending observations

Changes in airmass during a night are one source of systematic trends in ground-based photometry. Many stars observed in the same sequence show similar trends, allowing a common set of basis vectors to be derived and fitted to their photometry. We use the SYSREM algorithm (Tamuz et al. 2005), which has been used in wide-angle ground-based photometric surveys such as WASP and NGTS.

Photometry is first grouped by autoguider reference image, as each group has distinct offsets and systematics. We flag photometric data points affected by the VLT laser and those more than 7σ from the median flux for a given source. We perform a coarse decorrelation against airmass and CCD position to remove prominent trends, while down-weighting variable stars and images affected by clouds. We then use SYSREM to remove up to five of the most prominent common trends from the photometry of each target. Flagged data points do not contribute to the determination of the basis vectors. The detrended flux is stored in a two-dimensional FITS array of 32-bit floating-point numbers. A SYSREM summary file contains a FITS table in extension 1, recording the basis vectors, the stars used to derive them, the variance across each image (SIGMA_XS), and its photometric zero-point. The latter two quantities help identify measurements affected by clouds.

Testing shows that this approach is generally robust for photometry grouped by autoguider reference image, although individual nights occasionally remain poorly detrended for reasons that are not yet understood. We therefore also apply the coarse decorrelation and SYSREM separately to each action. This provides an alternative when detrending the full group is unsuccessful and may better capture trends specific to an individual night. However, detrending actions separately does not preserve photometric offsets between actions sharing the same reference image (see Section 5.3).

5.3 Aligning photometry from different cameras

Some targets are observed simultaneously with multiple telescopes, or with different telescopes at different stages of a photometric campaign. Differences in camera response introduce photometric zero-point offsets that must be accounted for. For multicamera transit observations, each camera's data set should be treated separately, with its own photometric zero-point fitted alongside the transit model. Monitoring campaigns that use different telescopes on different nights also require photometric offsets between the observations to be determined. In practice, a separate offset is required for each autoguider reference image, since offsets can arise even for observations from the same camera.[3]

The quick-look pipeline estimates photometric zero-point offsets for each autoguider reference image using median flux measurements for individual nights and the full data set. The reference image associated with the largest number of data points defines the zero-point to which the other groups are aligned. We first identify nights

[3] Autoguider reference images may change if the initial image contains clouds, laser contamination, or satellites, or following camera or telescope repairs.

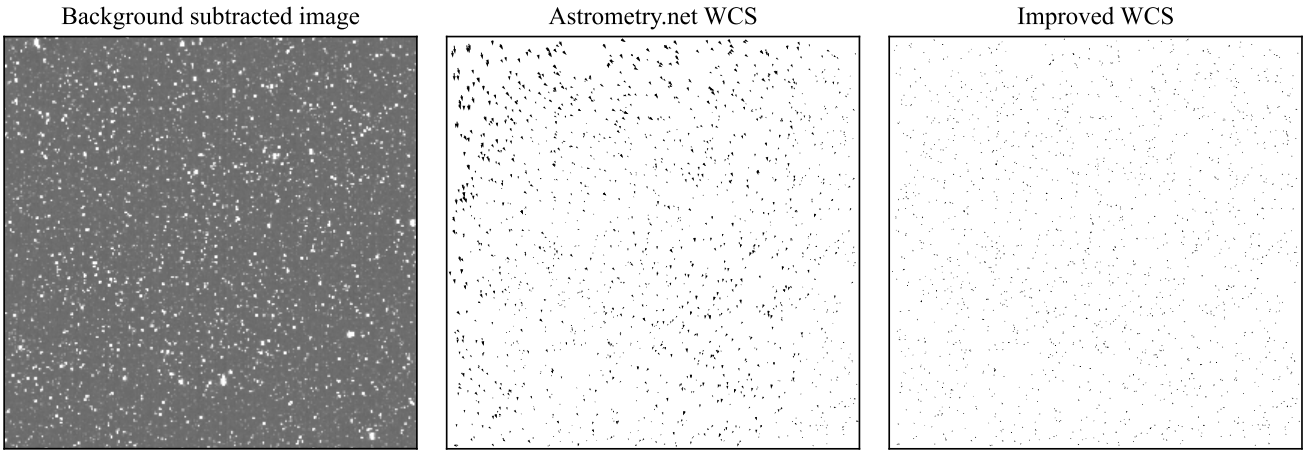


Figure 9. Left: raw image of a field centred at RA = $21^{h}\,41^{m}$, Dec = $-36°\,33'$. Centre: positional residual vectors for the ASTROMETRY.NET solution, using suitably bright, unblended sources. Right: residual vectors for the improved WCS described in Section 3.5. Both vector plots show only stars whose measured positions differ from the improved WCS predictions by no more than 0.1 pixels. Vector lengths are exaggerated by a factor of 100. The residual distributions for the full sample are shown in Fig. 10.

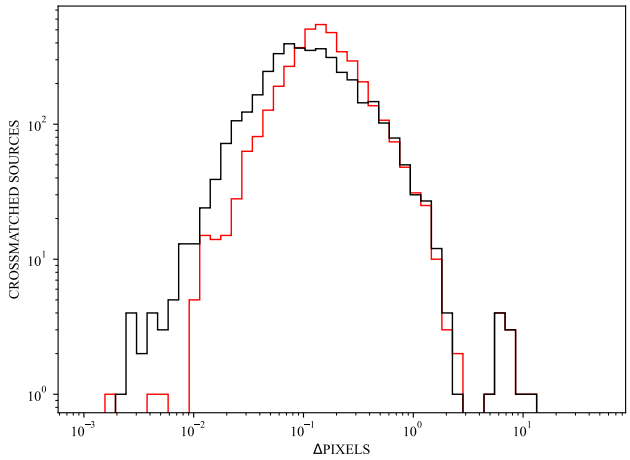


Figure 10. Positional offsets (in pixels) of cross-matched sources for the WCS solution calculated by ASTROMETRY.NET (red) and our refitted solution (black), for the NGTS image centred at RA = $21^{h}\,41^{m}$, Dec = $-36°\,33'$ shown in Fig. 9.

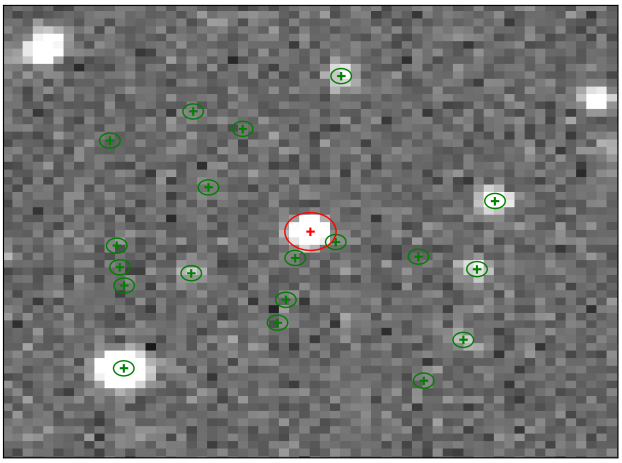


Figure 11. A subset of the target photometry catalogue for TIC 1676693, showing the target (red) and nearby stars that could contribute contaminating flux or false-positive signals (green).

with simultaneous observations in this group and groups associated with other reference images. Comparing the median detrended target fluxes yields offsets between the groups. We repeat this process until all groups are aligned or no further offsets can be determined from overlapping observations. For any remaining groups, we estimate offsets by comparing their median detrended target fluxes over the full data set with that of the reference group. These offsets should be used with care, and data associated with each autoguider reference image should be treated as a separate data set where possible.

5.4 Final product

In addition to the low-level per-action products (Section 5.1), the pipeline provides a final product as a single FITS file containing five table extensions.

Extension 1 contains the catalogue of sources for which photometry was extracted (Table 2), including a subset of columns from TIC V8.0. Three additional columns are provided: ra_deg_corr and dec_deg_corr give coordinates corrected for proper motion to the epoch of the autoguider reference image, while on_chip records whether the source falls on that image. During reduction, a more extensive catalogue, including sources with $T_{\rm mag} > 16$ and sources outside the reference image, is used to assist with WCS solutions. For the default magnitude limit, the final product contains sources

Table 2. Names, data types, and descriptions of columns in extension 1 of the final product (target photometry catalogue).

Column	Data type	Description
TIC_id	INT	TIC ID
Gaia_id	INT	*Gaia* ID
ra_deg	FLOAT	J2000 right ascension [degrees]
dec_deg	FLOAT	J2000 declination [degrees]
pmRA	FLOAT	Proper motion in right ascension [mas yr^{-1}]
pmDEC	FLOAT	Proper motion in declination [mas yr^{-1}]
i mag	FLOAT	Sloan *i* magnitude
GAIAmag	FLOAT	*Gaia* magnitude
Jmag	FLOAT	2MASS *J* magnitude
objType	STRING	Object type according to TIC V8.0
lumclass	STRING	Luminosity classification according to TIC V8.0
on_chip	INT	Flag indicating whether the source falls on the detector
ra_deg_corr	FLOAT	J2000 right ascension corrected for proper motion [degrees]
dec_deg_corr	FLOAT	J2000 declination corrected for proper motion [degrees]

Table 3. Names, data types, and descriptions of columns in extension 2 of the final product (observation summary log).

Column	Data type	Description
action_id	INT	Action ID
camera_id	INT	Camera ID
night	STR	Observing night
action_type	STR	Type of observation
actual_start_utc	DATETIME	Start of the observations
actual_end_utc	DATETIME	End of the observations
status	STR	Whether the action completed, failed to start, or was aborted
field	STR	Field name [NGXXXX-YY]
campaign	STR	Observing campaign name
autoguided	STR	Whether autoguiding was used
ref_image_id	INT	Reference image ID used for autoguiding
gain_idx	INT	Gain setting index
num_images	INT	Number of images obtained
mean_moon_dist	FLOAT	Mean angular separation from the Moon [degrees]
args	STR	Additional arguments for the observation
num_extracted	INT	Number of images reduced

on the detector (on_chip=1) with $T_{\rm mag} < 16$ for which photometry was extracted.

Extension 2 contains an observation summary log for the target. It records each action's camera ID, observing night, start and end times, completion status, reference image ID, camera gain setting, number of images obtained, and mean angular separation from the Moon in degrees. These columns are summarised in Table 3.

Extension 3 contains information from the SYSREM detrending of individual actions: the variance metric SIGMA_XS, correlation coefficients, and a flag indicating whether each star was used as a standard to derive the basis vectors. These three columns are repeated for each action and are summarised in Table 4.

Extension 4 contains information about individual images. Each row corresponds to an image and records its ID, flags set before reduction, observing night, airmass at the commanded right ascension and declination, telescope hour angle in degrees, and whether the image was reduced. These columns are summarised in Table 5.

Extension 5 contains photometry for a given target, including BJD, WCS and centroid positions, photometric flags, PSF metrics, flux measurements, and SYSREM outputs for individual actions and for all actions together. These columns are summarised in Table 6.

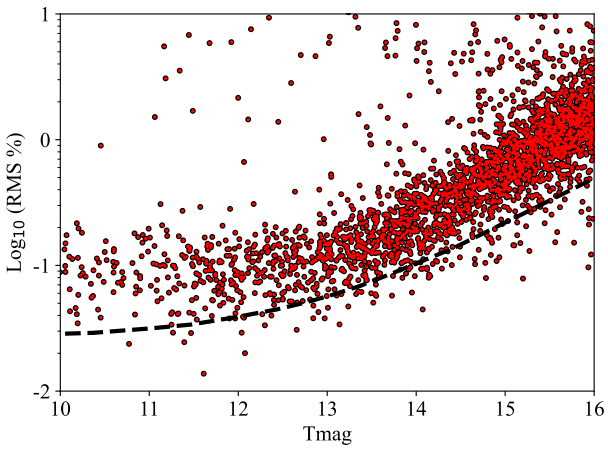


Figure 12. Photometric noise over 1 h for objects in a sample of 3130 sources observed with a single NGTS telescope at Paranal and reduced with the quick-look pipeline (red points with black outlines). We select targets not known to transit or eclipse and exclude multicamera observations. The schematic dark-sky I-band noise model ($I \approx T_{\rm mag}$) from Wheatley et al. (2018) is shown for reference (dashed black line).

6 PHOTOMETRIC PERFORMANCE AND EXAMPLES

Fig. 12 shows the photometric performance for 3130 objects observed with NGTS and processed with the quick-look pipeline. These focused observations were obtained to recover exoplanet transits for targets near the centre of the field. The measured noise is compared with the dark-sky noise model for a single NGTS telescope from Wheatley et al. (2018), illustrating the dependence of the achieved precision on target magnitude. The noise performance of defocused and multicamera observations has been tested elsewhere (e.g. Bryant et al. 2020a; O'Brien et al. 2022).

The photometric stability of NGTS is due in part to its autoguiding system. Reference images and the DONUTS algorithm (McCormac et al. 2013) allow each telescope to keep a target within the same pixel both within and between nights. The sources shown in Fig. 13 have positional scatter below 0.1 pixels in each coordinate. This reduces correlated noise associated with changes in detector position and supports stable photometry across multiple nights.

Examples of NGTS photometry appear in numerous publications.[4] Data reduced with the quick-look pipeline have supported photometric monitoring of planet hosts (e.g. WASP-39; Ahrer et al. 2023), precise multicamera transit observations (e.g. Bryant et al. 2020a; Armstrong et al. 2020; Barragán et al. 2022), and period recovery for *TESS* single-transit planets (Gill et al. 2020b). Observa-

[4] https://ngtransits.org

Table 4. Names, data types, and descriptions of columns in extension 3 of the final product (per-action SYSREM information).

Column	Data type	Description
SIGMA_XS_ACTIONXXXXXX	float	Variance metric for each measured source in action XXXXXX
CI_ACTIONXXXXXX	float	ASK RICHARD
IS_STD_ACTIONXXXXXX	int	Flag indicating whether a star was used as a standard to derive the SYSREM basis vectors

Table 5. Names, data types, and descriptions of columns in extension 4 of the final product (image information).

Column	Data type	Description
FLAGS	INT	Photometric flags set before reduction
IMAGE_ID	INT	Image ID
NIGHT	INT	Observing night represented as YYYYMMDD
AIRMASS	FLOAT	Airmass at the commanded right ascension and declination
TEL_HA	FLOAT	Hour angle at the commanded right ascension and declination [degrees]
reduced	bool	Boolean indicating whether the image was reduced

Table 6. Names, data types, and descriptions of columns in extension 5 of the final product (target photometry).

Column	Data type	Description
BJD	DOUBLE	Barycentric Julian date
BJD_SHORT	DOUBLE	Barycentric Julian date with gaps removed
X	DOUBLE	Target x-coordinate from the refined WCS
XC	DOUBLE	Target centroid x-coordinate from SEP
Y	DOUBLE	Target y-coordinate from the refined WCS
YC	DOUBLE	Target centroid y-coordinate from SEP
FLAGS_1	INT	Flags set by the per-action reduction
PFSSEP	DOUBLE	PSF width in the centre of the frame from SEP
PFSTOM	DOUBLE	PSF width in the central third of the frame, calculated from supersampled PSFs
PFSRICH	DOUBLE	PSF width calculated from a rolling stack of the last 10 frames
FLUX	DOUBLE	Target flux measured with the optimal aperture
FLUX_SYSREM	DOUBLE	Target flux detrended with SYSREM using all observations
FLUX_SYSREM_ACTION	DOUBLE	Target flux detrended with SYSREM for the action
FLUX_ERR	DOUBLE	Uncertainty on the target flux
FLUX_BKG	DOUBLE	Sky background flux
SIGMA_XS	FLOAT	Variance metric across the frame
ZERO_POINT	FLOAT	Photometric zero-point for the frame
AJ	float[]	ASK RICHARD
NIGHT_IDX	INT	Unique integer for each observing night
FLAGS_2	INT	Flags set after SYSREM
ZP_FIT	FLOAT	Fitted photometric zero-point
EXT_XSLOPE	FLOAT	ASK RICHARD
EXT_YSLOPE	FLOAT	ASK RICHARD
SIGMA_XS_ACTION	FLOAT	Per-action variance metric across the frame
ZERO_POINT_ACTION	FLOAT	Per-action photometric zero-point for the frame
AJ_ACTION	float[]	ASK RICHARD
NIGHT_IDX_ACTION	INT	Unique integer for each observing night within the action
FLAGS_ACTION	INT	Flags set after per-action SYSREM
ZP_FIT_ACTION	FLOAT	Fitted per-action photometric zero-point
EXT_XSLOPE_ACTION	FLOAT	ASK RICHARD
EXT_YSLOPE_ACTION	FLOAT	ASK RICHARD

tions can be requested for a variety of science programmes,[5] with data provided in a timely manner.

7 CONCLUSIONS

We have presented the high-precision targeted observing mode of NGTS and the quick-look pipeline used to reduce these observations. Targeted programmes include transit observations with one or more cameras (e.g. Bryant et al. 2020a; Armstrong et al. 2020; Barragán et al. 2022), monitoring to recover transits of long-period planets (e.g. Gill et al. 2020b, 2024), measurements of stellar activity (e.g. Ahrer et al. 2023), and coordinated observations with other facilities (Doyle et al. 2022, 2023).

The pipeline identifies pixels affected by VLT laser guide stars and satellite trails, estimates spatially varying sky backgrounds, and detects sources. Iterative cross-matching and refitting of the Simple Imaging Polynomial coefficients refine the WCS solution, yielding positional residuals of approximately 0.1 pixels for most matched

[5] Contact p.j.wheatley@warwick.ac.uk for more information.

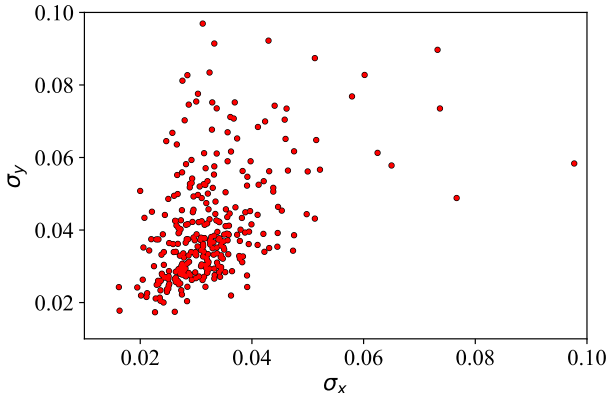


Figure 13. Tracking performance for objects observed with a single NGTS telescope at Paranal and reduced with the quick-look pipeline (red points with black outlines). Each axis shows the standard deviation of the target's pixel-coordinate offsets from its position in the reference image. We select targets not known to transit or eclipse and exclude multicamera observations.

sources. Catalogue-based aperture photometry is accompanied by measurements of the point-spread function and source centroids, together with barycentric time corrections and observing diagnostics. Common systematic trends are removed using SYSREM, and photometric zero-point offsets between data sets are estimated when observations from different cameras or reference images are combined.

The pipeline provides both low-level per-action products and a consolidated FITS file containing the source catalogue, observing logs, diagnostic information, and photometry. The observing strategy, instrumental stability, and reduction procedures together produce publication-quality light curves with precision comparable to the NGTS noise model (Section 6). Processing at Paranal enables these products to support timely follow-up of exoplanet candidates and coordinated observations with other facilities.

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DATA AVAILABILITY

The data used in this work are available from the corresponding author upon request.

[6] https://www.astropy.org

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